

Capps, President of Epic Games gives it a shot—“The answer is the *Unreal 3* engine”. But that was used for the original game too, our plaintive protest. What’s new? “You want it, you got it!”—with those ominous words hanging in the air Mike rattles off about the various technical aspects of *GOW 2*. Here are the words straight from the man’s mouth:

1. Ambient Occlusion / Post-Processing Filter: *Unreal Engine 3* (UE3)’s ambient occlusion technology renders an approximation of global illumination by calculating the accessibility of each pixel. The result is increased perception of geometric shape for the entire scene, including dynamic characters. The effect is scene-independent and operates with constant memory and performance overhead with no pre-processing.

2. Advanced Character Lighting: Improvements to the UE3 character lighting system enable the engine to more accurately synthesise complex light environments. Fill lighting is added using the increased accuracy to maintain the contrast of character details. Character shadowing accuracy has also been enhanced.

3. High-density Crowds: UE3’s new crowd system with flocking technology can simulate in real-time hundreds of characters within a scene. Battling the Locust Horde in the original *Gears Of War* often involved attacking five or six enemies at once, and it’s now possible to have masses of enemies onscreen at once.

4. Dynamic Fluid Surfaces: The water technology in UE3 breathes life into liquid surfaces. Sophisticated water physics combined with realistic specular and environment reflections produce a realistic, liquid-like feel. Characters have dynamic interactions with the water and this adds a layer of realism.

5. Matinee improvements: Matinee is used to build real-time cinematics that play back in-engine. This UE3 system has been substantially improved to give artists movie director-class control over all of the objects in a scene, as well as cameras and cuts, plus we have added an in-engine, real-time preview feature that plays back all the visual effects of a cinematic.

6. Soft body physics: Through the incorporation of Aegia’s soft body physics simulation technology, UE3 can realistically simulate elastic and deformable objects, and thus produce more organic game worlds. Players can interact with flexible objects, and gelatinous surfaces can exhibit sticky properties.

7. Destructible environments: UE3’s new fracturing tool and, enable developers to take virtually any existing mesh, slice it up into as many fragments as desired, and destroy structures. This structural analysis tool produces far more realistic, destructible environments.

He continues: “Many gamers still consider the original *Gears of War* to be the technical bar for next-generation gaming—and that was our first Unreal Engine 3 title. We’ve been working on the technology for two years, so of course, there are tons of great new features. It’s also been greatly optimised, which helps us to make *Gears of War 2* even bigger and badder than the first game”.

After that rather lengthy description and a look at some of the screenshots, and in game footage of *GOW 2*, it is time for a little break, hosted by Cliff B, as we’re led through the corridors of the building for the second time. We’re at the front lobby now and the time is around 1 pm. Several employees are just entering the building. Cliff reiterates the casual work timings, “They leave late, (incidentally, 7 pm is late for them!), so they’re in late as well. The core working hours are 1.30 pm to 5.30 pm, and we try to ensure our presence during these hours, otherwise it’s mostly cool”. The back of the building has a rather large open-air veranda with a few chairs and tables spread around. Someone asks what is perhaps the most clichéd question ever—what are the three best things about working here?

Cliff replies: “One, the people... I work with the most intelligent and talented people in entertainment, and every day is filled with challenges and surprises that keep me on my toes. Two, the environment... our office feels professional, yet cosy. There are no cubicles, and food is provided during the busier times, and everyone’s got action figure and lastly creative freedom. Lastly, Epic builds our own universes, and thus, we have enough wiggle room to dream big”.

A couple of questions more about Epic’s prosperity after their recent success with *GOW* and the just released *Unreal Tournament 3* had Cliff joking “Our car park has been upgraded quite a lot over the past two years, with many faster, more expensive cars”. We share a moment of mirth and head back inside. *En route* their press room, we make a quick stop at Cliff’s office. His desk itself is huge, strewn with magazines, odds and ends, and to top it all, a multitude of action figures—Cliff is a proud collector of more than 150 such action figures. A pretty impressive looking PC with a large monitor dominates one side of the desk and the action figures take up another third, along with tons of images depicting concept art from titles he’s worked on, including *Unreal Tournament 2003*, *UT 2004*, and *Gears of*

War. His desk is his den, and obviously the heart of his productive soul.

After a little more talk about *GOW 2*, and the technical aspects of the *Unreal 3* engine, we went for a late lunch. During the healthy yet sumptuous meal, we got talking again:

Does game development follow a particular procedure? Are there any stages to the procedure? If so, what are they?

Mike Capps: There most certainly are procedures and stages to the process. It’s like creating any other product. There’s a lot of hard work, and most of us put in very late hours, but since we all love what we do, we have loads of fun too. Game development is a complex and expensive process. Most ideas never become reality because today’s top-tier games cost tens of millions of dollars (and years!) to develop, and the respective marketing budgets are just as costly.

OK, so what about those stages?

Capps: Ah yes! For us, the first stage is the pitch. Epic continuously canvasses its employees for fun and original game concepts, and these ideas require extensive evaluation. These concepts are narrowed down through discussions, in which everyone is allowed to participate. The submissions are then presented for a more formal evaluation for key stakeholders. If the concept is scrapped at this stage, we start from scratch and the process mentioned above is repeated. If a concept is greenlit, or approved for production, the game-to-be moves into pre-production. Everyone has to be totally committed to a concept before it gets greenlit for production.

During pre-production, (Stage 2, in-case you were counting!), producers and designers create design documents and technical specifications based on the development timeline. It’s essential for large teams to assess the myriad needs during these early phases in order to maintain quality and keep games on schedule.

At this point, we start getting the idea that game development isn’t all fun and (pardon the pun!) games, and there seems to be a whole lot of work involved, with close teamwork and coordination between teams. We questioned Mike a bit more on what a design document is.

Capps: Design Documents are sort of a wish list for components and essential elements that the game should contain or implement or include. Technical specifications include the nature of the engine, the special effects, physics and other specifics like shading, lighting and damage models to be used.

The next stage involves game production. Full-on game production involves implementing features described in the design document, and it’s a constantly evolving process as

features may be added, dropped or modified depending on time or technical constraints.

The production stage can be broken down into phases called alpha, the first playable version of a game, and beta, the playable version complete with features and assets.

Capps adds that Epic’s motto is polish, polish and polish. To further quote him “When a game reaches beta, it may contain bugs or gameplay balancing issues, so we spend a lot of time tweaking gameplay and optimising levels to satisfy internal expectations and standards. Quality assurance testing is a challenging process that can last six months or more. Our testers bang on our games, time and again to ensure that we have explored every possible decision and taken every path in-game to ensure a smooth and exciting user experience”. This is an important stage for a thoroughly professional outfit like Epic, especially considering the scale of the games they work on, and even a single overlooked aspect could cause a lot of problems. We’ve all played games, which have certain levels beyond which progress is impossible due to a bug, or a game crash, or a part of a level that may be inaccessible, and we know how annoying that can be.

Whew! Is that all? We ask hopefully. Not quite, Mike quips—there’s Post Production which wraps up with the end of QA testing, submission to console certification when appropriate, and approval for manufacturing. Once this transpires, the game is sent to the manufacturer for replication, and packaging materials are printed as well. Components are then assembled, packed, sealed and shipped to retailers.

Then there’re a lot of other concurrent processes in addition to game development. While making a game, there are teams of external



Entertainment galore...Epic plays as hard as they play...err work



Typical Epic productivity dens



A few of the Epic team caught just after we emerged from our hands-on with *GOW 2*



A typical designers desk is never complete without action figures